Supplementary Material for Descriptor: Underwater Stereo Vision Dataset for Transparent Object Detection & Ranging (UW-TransStereo)

Supplementary Figures

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

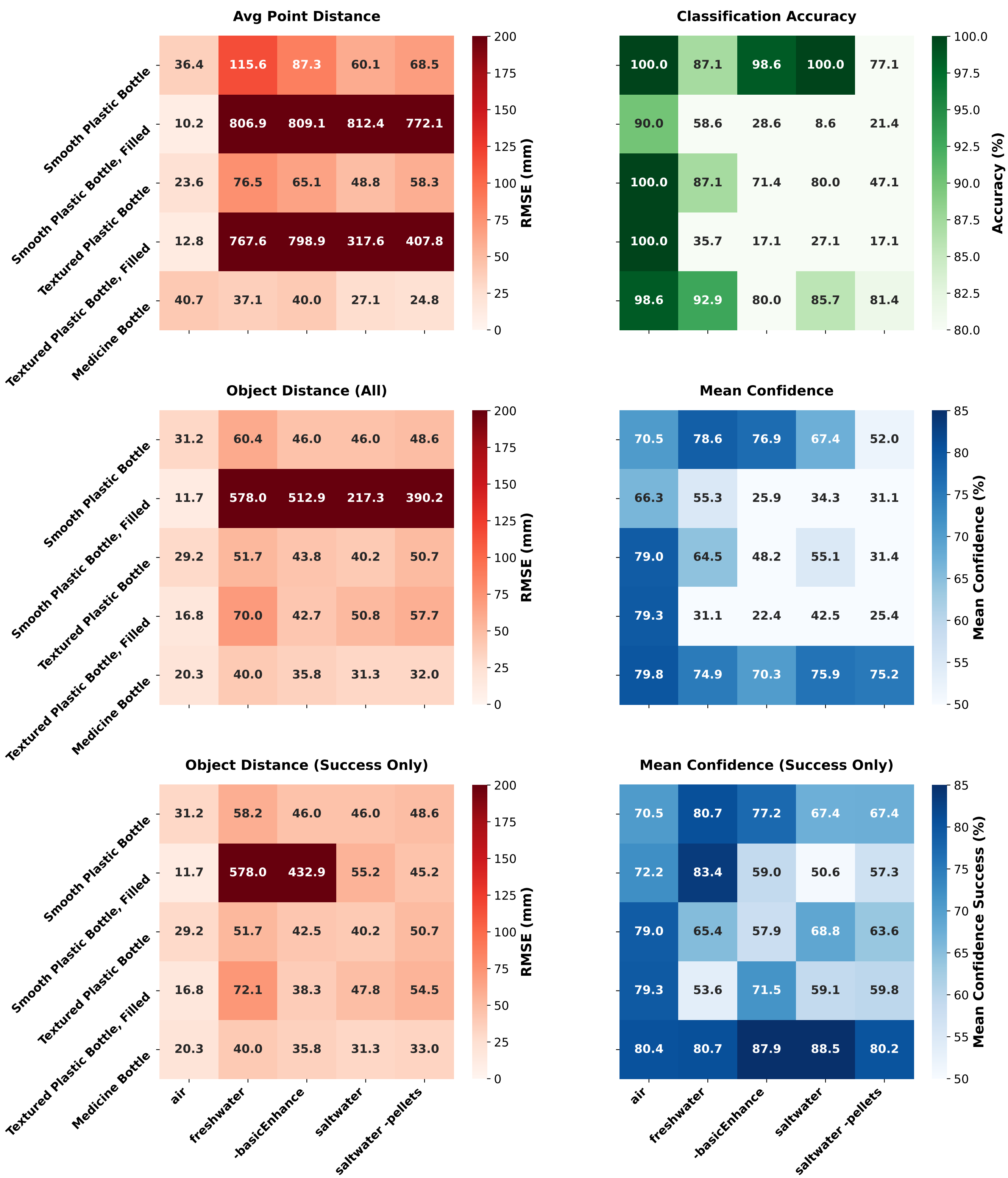

FIGURE S1. Performance metrics across environments (x-axis) and bottle types (y-axis) in a 3×2 layout on distance-stratified normalised data (10 samples per bottle type per distance, 200–800 mm range, includes failed detections as zeros). Left column shows root mean squared error (RMSE) for distance-based metrics: (A) Avg Point Distance, (B) Object Distance (All detections), and (C) Object Distance (Successful detections only). Right column shows classification performance: (D) Classification Accuracy, (E) Mean Confidence (all detections), and (F) Mean Confidence (success only). Distance RMSE values are in cm; classification accuracy is a percentage; confidence values are YOLO probabilities in [0,1] shown as percentages.

Supplementary Tables

TABLE S1. Archive manifest (excerpt example).

Path/Filename convention (relative)	Type	Purpose
Air/medBtl/recording_07052025_234742.svo2	SVO2	Raw rectified stereo
Air/medBtl/detections_07052025_234742.xlsx	XLSX	Per-frame detections
Air/medBtl/screenshot_*.png	PNG	Qualitative screenshots
Freshwater/medBtl/recording_08052025_020841.svo2	SVO2	Raw rectified stereo
Freshwater/medBtl/detections_08052025_020841.xlsx	XLSX	Per-frame detections
Freshwater/medBtl/screenshot_*.png	PNG	Qualitative screenshots
Saltwater/medBtl/recording_08052025_182447.svo2	SVO2	Raw rectified stereo
Saltwater/medBtl/detections_08052025_182447.xlsx	XLSX	Per-frame detections
Saltwater/medBtl/screenshot_*.png	PNG	Qualitative screenshots
Saltwater_pellets/medBtl/recording_08052025_191235.svo2	SVO2	Raw rectified stereo
Saltwater_pellets/medBtl/detections_08052025_191235.xlsx	XLSX	Per-frame detections
Saltwater_pellets/medBtl/screenshot_*.png	PNG	Qualitative screenshots

TABLE S2. Selected columns in collated CSVs.

Column	Unit	Description / Policy
environment	–	{Air, Freshwater, Saltwater, Saltwater_pellets}
bottle_type	–	{medBtl, smtBtl, texBtl} with optional -fill suffix
class	–	COCO detection class ID (39 = bottle)
name	–	Detection class name (bottle, person, vase, etc.)
confidence	[0,1]	Detector confidence score (parsed from percentage)
center_distance	cm	Stereo range at center ROI; NaN if unranged
upper_distance	cm	Stereo range at upper ROI; NaN if unranged
lower_distance	cm	Stereo range at lower ROI; NaN if unranged
object_distance	cm	Calculated object distance from stereo
width	cm	3D object width measurement
height	cm	3D object height measurement
depth	cm	3D object depth measurement
GT_*	–	Ground truth data
timestamp	–	Recording time reference (ISO format)